Independent Landsat analysis of river-avulsion disturbance

A partial reproduction of Valenza et al. (2020)

Independent computational reconstruction

12 October 2026

Abstract

River avulsion can expose sediment and water outside the preceding channel. Valenza et al. (2020) infer avulsion style from the area of this disturbance, normalized by parent-channel area. I implement the written spectral and statistical methods in C++17 and apply them to independently obtained USGS Landsat observations. The geographic distribution of all 63 permitted site locations is reconstructed. Four events are investigated with annual imagery; an additional, larger domain is acquired for one of them. B12 and Sarare V11 footprints are calculated over explicitly incomplete spatial and temporal windows. The calculations pass analytical and raster consistency checks, but the 63-event geometry and disturbance dataset is not independently recovered. Consequently, the central downstream correlations and morphology distributions are not reproduced. This report preserves the generated observations, missing results, and the commands needed to repeat the attempt.

1 Scope and evidential boundary

This is a partial observational reconstruction, not a forward simulation of avulsion. The paper does not prescribe evolution equations that predict channel relocation from initial and boundary conditions. Its quantitative observable depends on satellite measurements and interpreted event boundaries [1]. Implementing spectral arithmetic alone does not recover those boundaries or establish the inferred sedimentary mechanism.

No author code, plotting script, processed image, channel mask, source measurement table, or digitized plot enters the analysis. The user permitted only published site identifiers and coordinates to locate the observations. `config/sites.csv` contains those 63 records. An optional extractor reads only worksheet columns A–C; it retains no event dates or measured outcomes. The paper and its written supplementary methods supply the scientific specification. The two expressly permitted project-guidance files are archived under `guidance/`. No other content of the guidance repository is used.

During an earlier feasibility discussion, the assistant inspected the supplementary measurement workbook. That prior exposure cannot be undone. The implementation excludes those measurements and discloses the exposure in `docs/provenance-policy.md`; this is not a blinded clean-room claim. Every image in this report was generated by this project. No published reference image is embedded. Missing-panel notices are accounting devices, not data or successful reproductions.

2 Independent observations and methods

2.1 Acquisition and compositing

The primary observations are USGS Landsat Collection 2 Level 2 surface reflectance and quality bands, obtained through the public Planetary Computer catalog. They are agency products, not the paper's processed satellite images. Atmospheric correction performed by the provider is accepted, as in the paper's use of surface reflectance. The selected missions are Landsat 5, 7, and 8; the principal archive interval is 1986–2017. Four approximately 20-km domains surround B11, B12, V11, and V15. A roughly 60-km B12 domain extends northeast along a channel observed in our own imagery. Its selected years are 1995 and 2010–2017. Bounds, UTM coordinate systems, and sampling are explicit in `config/events.json` and `config/extended.json`.

The acquisition script range-reads the provider's cloud-optimized GeoTIFFs and places eight unsigned-integer bands on fixed 30-m grids with nearest neighbor resampling. These are blue, green, red, near infrared, two shortwave infrared bands, QA_PIXEL, and QA_RADSAT. Unsigned source URLs and frozen scene identifiers are recorded in `data/manifests/`; downloaded array hashes and failures are in `data/provenance/`. Access tokens are transient and are not archived.

The C++ program converts nonzero digital numbers to reflectance using the USGS Collection 2 relation [3],

$$\rho_b = 2.75 \times 10^{-5}\,\mathrm{DN}_b - 0.2. \tag{1}$$

Zero digital numbers remain missing; reflectance is not clipped. A sample is excluded when a required band is missing, QA_RADSAT is nonzero, or QA_PIXEL flags fill, dilated cloud, cloud, cloud shadow, snow, or Landsat-8 cirrus [4]. This documented QA screen is an adaptation: it is not the paper's GEE simple-cloud-score threshold of 20%. The retired Collection 1 archive and unspecified additional radiometric alignment cannot be exactly reconstructed from the supplied written description [5].

For each year, the implementation selects the sensor with the largest number of clear pixel-observations in the domain, breaking a tie in favor of the larger sensor number. This explicit assumption avoids applying one sensor's coefficients to another sensor's band median. It also discards otherwise usable observations from other sensors that year. The paper does not fully specify mixed-sensor compositing or alignment. Annual composites use the per-band median of retained observations at each pixel; pixels with no observation remain missing. Even-sized samples use the mean of the central two values. Missing years are not interpolated.

2.2 Spectral fingerprints

For each annual composite, the tasseled-cap transform is

$$T_j(\mathbf{x}, y) = \sum_{b=1}^{6} C_{jb}\ \mathrm{median}_{t \in y}\,\rho_b(\mathbf{x}, t), \qquad j \in \{B, G, W\}. \tag{2}$$

Here $\mathbf{x}$ is a pixel location, y is calendar year, and B, G, and W denote brightness, greenness, and wetness. The following coefficients are the written Supplementary Table 2 specification [2]; they are not fitted to any published outcome. Physical-band order is blue, green, red, near infrared, SWIR1, SWIR2.

Sensor	Component	Blue	Green	Red	NIR	SWIR1	SWIR2
TM	B	.2909	.2493	.4806	.5568	.4438	.1706
	G	-.2728	-.2174	-.5508	.7221	.0733	-.1648
	W	.1446	.1761	.3322	.3396	-.6210	-.4186
ETM+	B	.3561	.3972	.3904	.6966	.2286	.1596
	G	-.3344	-.3544	-.4556	.6966	-.0242	-.2630
	W	.2626	.2141	.0926	.0656	-.7629	-.5388
OLI	B	.3029	.2786	.4733	.5599	.5080	.1872
	G	-.2941	-.2430	-.5424	.7276	.0713	-.1608
	W	.1511	.1973	.3283	.3407	-.7117	-.4559

The supplementary table repeats TM band labels for OLI. I interpret its coefficients as physical spectral bands, mapping OLI bands 2–7, rather than including OLI's coastal band. This is an explicit resolution of a labeling ambiguity.

Let $Q_p(T_j)$ be the p quantile over valid pixels in an annual domain. The two candidate signatures are

$$I_B = [T_B > Q_{0.90}(T_B)]\,[T_G < Q_{0.50}(T_G)], \tag{3}$$

$$I_W = [T_W > Q_{0.75}(T_W)]\,[T_G < Q_{0.50}(T_G)]. \tag{4}$$

Square brackets are indicator functions. Quantiles linearly interpolate at sorted index $(n-1)p$, with strict inequalities at thresholds. This convention is not specified in the paper and is declared here. The rectangular domain also defines the percentile population; changing its extent can change detections even where the imagery is identical.

A mask M_A isolates the independently interpreted avulsion, and M_P isolates the preceding parent channel. Excluded persistent water and unrelated disturbance belong outside both masks. For a selected signature I, the duration fingerprint, occupied area, and style ratio are

$$F_A(\mathbf{x}) = M_A(\mathbf{x}) \sum_{y=y_0}^{y_1} I(\mathbf{x}, y), \tag{5}$$

$$A_A = (900\ \mathrm{m}^2) \sum_{\mathbf{x}} [F_A(\mathbf{x}) > 0], \qquad S_R = A_A / A_P. \tag{6}$$

The parent area uses its own comparable observation window. Repeated detection increases duration, not occupied area. A zero parent count produces an undefined ratio. Both signatures are preserved; the smaller avulsion area selects the signature, with brightness used in an exact tie. Selecting the smaller ratio would be a different operation.

Published masks are excluded. Independent polygons are specified by pixel-edge vertices; inclusion is tested at pixel centers. Annual observations are retained for audit even when unusable for event interpretation. A complete event analysis must separately reject inadequate years. No incomplete composite is silently filled or interpreted as an absence of disturbance.

2.3 Channel measurements and statistical operations

The normalized downstream coordinate is D/w, with centerline distance D from the mountain front and channel width w in meters. Local slope is $(z_u - z_d)/L$, where L follows the channel-belt reach. The paper uses kilometer-scale elevation averaging and roughly 5–10-km reaches. A coarse regional terrain map is not a substitute for that measurement. Flow-thread counts at several cross sections distinguish braided, transitional, and single-threaded channels. The written methods do not fully prescribe the cross sections or classification cutoffs.

The implemented statistical stage reads independent channel measurements, combines them with measured event ratios, calculates quartiles and full-range whiskers, and fits a line in $\log_{10}(S_R)$ versus $\log_{10}(D/w)$ for $D/w > 60$. Log-space ordinary least squares is an explicit interpretation of the described best-fit relation. No fit is emitted with fewer than three points or zero variance in either transformed variable. The current channel-measurement table is empty. No morphology labels, widths, slopes, or mountain-front distances are inferred from published outcomes.

The missing manual observations are material, not a software dependency that can be resolved by substituting values. Only four of 63 events were acquired and inspected in this attempt. The other 59 have location records but no independently processed local imagery or channel geometry. All 63 remain without a complete independent event-and-geometry record.

3 Implementation and verification

`src/avulsion.cpp` implements cloud screening, calibration, medians, transforms, thresholds, polygon counting, duration sums, ratios, and the statistical summaries. CMake builds a C++17 executable with optional OpenMP. Python handles provider I/O, reprojection, plots, archiving, and report assembly. Plotting reads saved C++ outputs; it never invents missing data or reads a manuscript figure. No neural image generator or conceptual river simulation is involved.

Annual processing stores one year's selected-sensor observations and uses selection algorithms for medians and quantiles. Expected spectral work is linear in pixel count, number of scenes, and band count. Pixel calculations run independently across OpenMP workers; temporal sums and output ordering are deterministic. Memory grows with one year's raster stack, rather than the full multidecadal archive. The largest domain is 1993 by 2055 pixels. The documented run uses 16 OpenMP threads.

Analytical checks include missing-data limits, strict thresholds on a constant scene, reflectance calibration, unit spectra, equal-area $S_R = 1$, zero-area limits, invariance of area ratios under common scaling, exact polygon area, and the distinction between temporal union and duration. The integration fixture sends known spectra, a cloud-contaminated observation, and masks through the executable. A synthetic $S_R = (D/w/100)^2$ relation tests regression and group statistics. These synthetic inputs exist only under `.runtime/`; they never supply a manuscript panel.

Check suite	Checks	Passed
validation	75	75
integration_checks	14	14
real_checks	401	401

Real-data audits enforce coverage and percentile-derived area bounds, compare saved duration sums against annual masks, and check that detection counts do not exceed valid observation years. Lossless array archives are also read back and compared element by element, preserving NaNs. These tests verify numerical operations. They do not validate cloud attribution, avulsion boundaries, sediment volumes, or the paper's scientific conclusion.

4 Generated observations and comparison

4.1 Geography and coverage

Figure 1 reconstructs the site distribution from the permitted coordinates and independently obtained terrain. Counts are 16 northern Andean, 14 central Andean, 6 Himalayan, and 27 New Guinean events. All 63 coordinate pairs are preserved by the C++ catalog calculation; the maximum coordinate residual at the stored precision is zero. This comparison checks a permitted input transformation, not independent discovery of sites. B15 appears here although the paper's central-Andes map omits that southern site. Map bounds, terrain sampling, and label placement differ.

Domain	Scenes	Years	Min. coverage	Median coverage	Max. coverage
B11	923	32	0.966	1.000	1.000
B12	1837	32	0.993	1.000	1.000
V11	765	27	0.001	1.000	1.000
V15	572	27	0.009	1.000	1.000
B12_wide	720	9	1.000	1.000	1.000

Coverage is the fraction of domain pixels with at least one valid annual reflectance sample. Years without any catalog scene are absent, not zero-filled. The two Sarare domains lack 1988 and 1992–1995 observations in the retrieved catalog; several other early annual composites have poor clear coverage. Figure 2 exposes these gaps. A first reconnaissance limited the input to two or four scenes per year, selected by scene-wide cloud cover. It often selected adjacent-path images that missed the site and retained Landsat-7 scan-line gaps. The final acquisition uses all selected catalog scenes. Choosing the annual sensor by clear pixel-observations also replaces an initial choice by scene count. The discarded scene-cap analysis is recorded as reconnaissance; it is not used as a final fingerprint.

4.2 The Sarare observation and its limits

The V11 sequence shows a southern meandering course in 1986 and a northern course with adjoining exposed surfaces in the 1990s. Parts of the northern course are already visible in the partly cloudy 1991 composite; a complete course is clear by 1996. The available sequence does not resolve initiation or completion. The broad disturbed corridor also intersects the eastern domain boundary. I therefore retain a *partial* 1996–1998 footprint, with a separate 1986 parent observation. Its vertices, inspection record, and limitations are in `config/annotations/V11/`.

Signature	Parent area (km^2)	Partial event area (km^2)	Area ratio
B	0.3717	2.7450	7.3850
W	2.2014	22.8141	10.3635

The brightness signature selects the smaller event footprint. Its area ratio is approximately 7.38, whereas the paper reports $S_R = 16.0$ for V11 in its discussion of Fig. 5 and labels Fig. 5e with $S_R = 14.9$ [1]. Numerically, the diagnostic is about 46.2% and 49.6% of these two reference values, respectively. The publication's discrepancy is not resolved here. These are not commensurate event estimates: our image domain truncates the event, the windows have unequal duration, the masks are approximate, and the spectral processing differs. No recovery of the published S_R is claimed. The wetness signature yields a much larger partial disturbance area, demonstrating sensitivity that a single selected number would conceal.

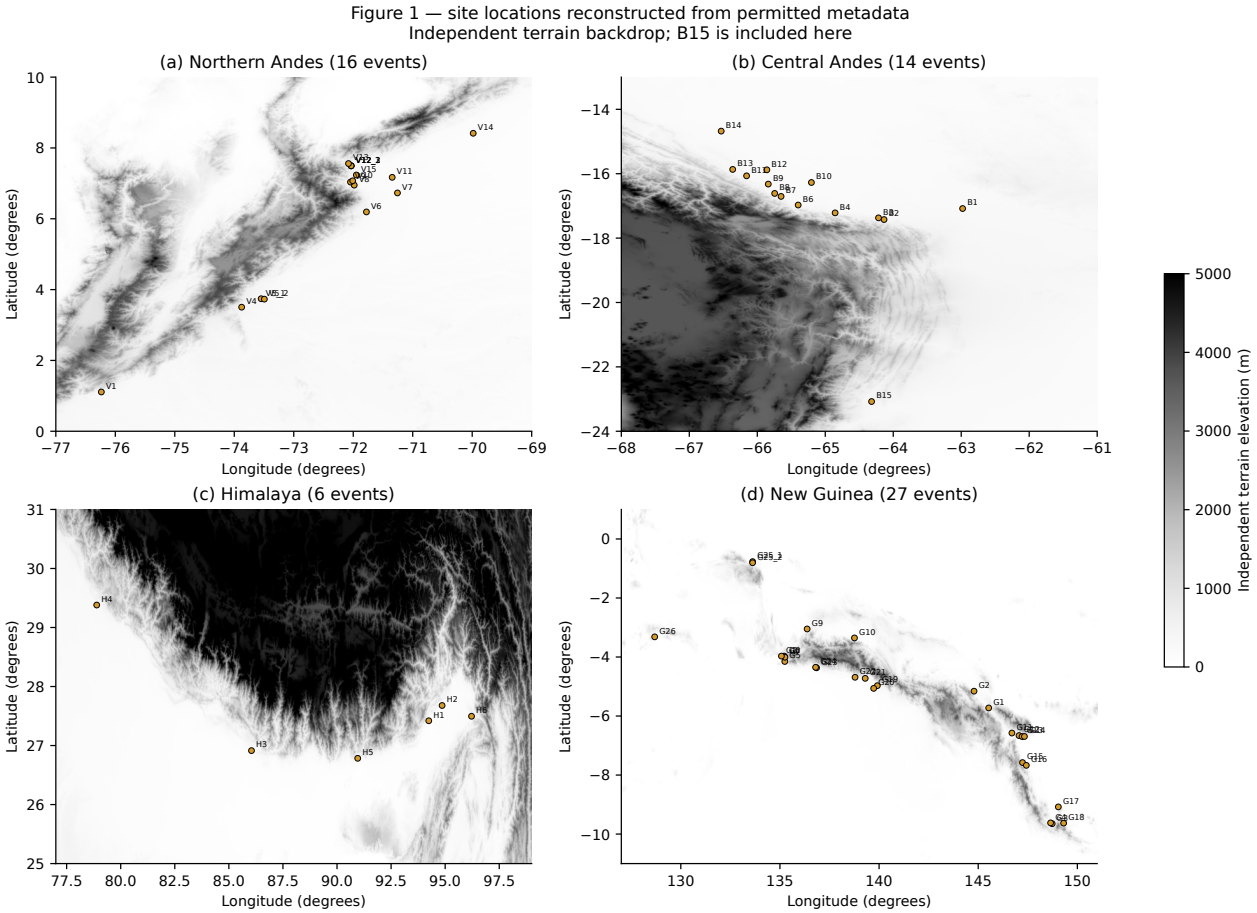


Figure 1: Independent reconstruction corresponding to manuscript Fig. 1a–d. Points are the permitted site metadata processed by the C++ catalog command. Shading is independently resampled global Terrain Tiles elevation in meters; it is not the paper's image or a measurement of local channel slope. Site distribution is reproduced; cartographic styling and terrain resolution differ. SRTM and GMTED2010 data are courtesy of USGS; ETOPO1 data are from NOAA/NCEI.

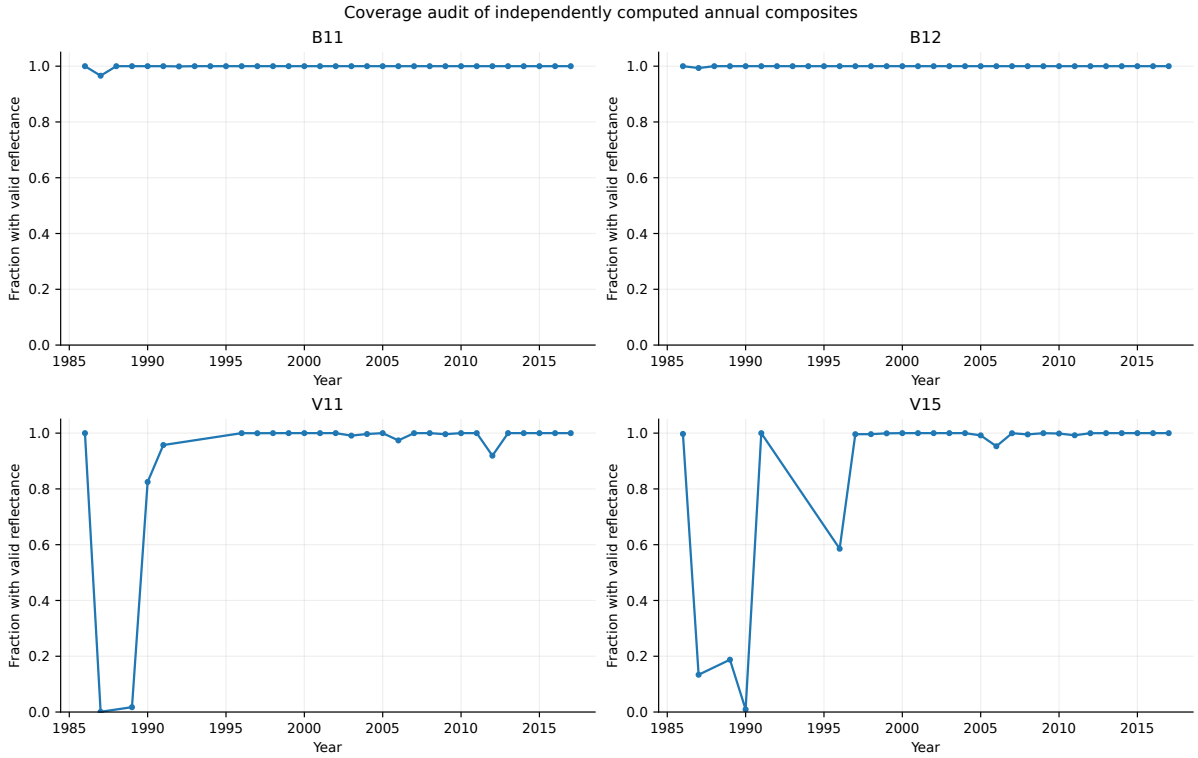


Figure 2: Coverage of the independently calculated 30-m annual composites in the four approximately 20-km domains. Gaps between years are shown by connecting lines for readability, not temporal interpolation. Low coverage prevents an absence of detections from establishing an absence of avulsion activity.

The V15 imagery shows a braided belt, but this attempt does not uniquely separate one avulsion from ordinary braid migration. A V15 event mask is therefore absent. The near-site context is independently generated, while the corresponding fingerprint panel remains unreproduced. A straight-line connection between V11 and V15 would not measure their along-channel separation; only their positions are shown in the overview attempt.

4.3 B12, B11, and the statistical panels

At B12 the independently acquired annual composites show changes in channel occupation. The larger domain follows the observed channel northeast and retains the years used by Supplementary Fig. 1. The partial central-reach footprint uses an independently drawn polygon and the 2010–2017 observation window; its parent is a truncated old-channel segment observed in 1995. The northeastern polygon boundary crosses the continuing channel, so its area cannot represent the complete event. This supplies partial Fig. 2a,b and four annual counterparts in Fig. 2c. Exact spatial agreement with the published central splay crop cannot be established from a site coordinate alone. At B11, narrow channels and uncertain event attribution prevent a defensible complete fingerprint. Candidate spectral detections are preserved as diagnostics, but they are not substitutes for avulsion masks.

Signature	Parent area (km^2)	Partial event area (km^2)	Area ratio
B	0.6426	15.0012	23.3445
W	3.7080	17.6508	4.7602

The B12 table reports the partial central reach only. Brightness selects the smaller avulsion area even though its ratio is larger than the wetness ratio. This illustrates why the stated selection rule must operate on avulsion area, not on the desired value of S_R. Neither ratio is used in the statistical figures or compared with an author measurement table.

Post hoc inspection of manuscript Figs. 2 and 5 was restricted to assessing agreement after these masks and windows were fixed. It exposes a material B12 mismatch: the publication displays the parent above the disturbed avulsion reach, whereas our limited sequence identifies an older southern course and a later northern course. The reference has no georeferenced axes, so display orientation alone cannot establish a north/south reversal. Orientation, a different relocation phase, or incorrect event attribution could explain the unmatched layouts. The present data do not resolve it. Thus the B12 panels are partial spectral-method attempts, not a verified reconstruction of the publication's event or spatial progression. No input polygon was adjusted to resemble the reference.

The published Fig. 2 duration legend reaches 20 years; our selected window contains eight. The reference's annual counterparts are 2007, 2010, 2013, and 2017, whereas the previously declared attempt years are 2011, 2013, 2015, and 2017. Those attempt years remain unchanged. The reference validation dates are 30 November 2007, 2 November 2010, 19 August 2013, and 30 August 2017; they are comparison information only. Their commercial images were not independently acquired. The V11 reference also includes extensive southern disturbance omitted by our northern polygon. These comparisons support the partial status rather than numerical validation. The assessment is archived in `docs/published-comparison.md`.

4.4 Sensitivity to the analysis domain

The nested B12 domains permit a numerical comparison over identical physical pixels. The C++ comparison maps their aligned 30-m grids, retains pixels valid in both composites, and reports

foreground intersection-over-union and total pixel agreement for each spectral signature. Mean absolute RGB difference is in reflectance units. Zero difference would establish agreement of the visible bands only; the other spectral bands and transforms would still need checking. Nonzero differences can reflect sensor selection, scene composition, or reprojection. This is a combined domain-sensitivity assessment, not an isolated threshold experiment.

Year	Signature	Intersection/union	Pixel agreement	Mean RGB difference
1995	B	0.569	0.993	9.862e-05
1995	W	0.242	0.906	9.862e-05
2010	B	0.517	0.985	3.853e-05
2010	W	0.175	0.946	3.853e-05
2011	B	0.510	0.991	6.038e-05
2011	W	0.121	0.861	6.038e-05
2012	B	0.381	0.994	4.921e-05
2012	W	0.247	0.846	4.921e-05
2013	B	0.273	0.981	0.002775
2013	W	0.163	0.947	0.002775
2014	B	0.610	0.992	4.84e-05
2014	W	0.275	0.960	4.84e-05
2015	B	0.429	0.990	5.193e-05
2015	W	0.282	0.965	5.193e-05
2016	B	0.390	0.978	4.298e-05
2016	W	0.288	0.968	4.298e-05
2017	B	0.152	0.977	0.003954
2017	W	0.214	0.875	0.003954

Overall agreement can be dominated by background pixels, so it is reported alongside foreground overlap. This comparison is an internal sensitivity assessment, not agreement with a published mask. It demonstrates why a geographic site coordinate alone does not fully specify a fingerprint. The command is `build/avulsion compare-domains data/derived/B12 data/derived/B12_wide data/results/domain_sensitivity.csv`.

The independently acquired Landsat data have 30-m pixels, each representing 900 m^2. A three-pixel boundary uncertainty is already 90 m. The commercial WorldView validation images in manuscript Fig. 2c and Supplementary Fig. 2c were not independently acquired. Consequently, this project cannot calculate pixel precision, recall, intersection-over-union, or progradation-front error against those observations. No published image is digitized to manufacture such scores.

Figures 3a,b, 4, and S3a,b remain unreproduced. The arithmetic for their quantities is implemented and tested, but no complete independent event and channel-geometry record supplies their plots. Empty CSVs and explicit absence notices are preserved. A 63-point regression, median morphology contrast, or significance claim would require measurements that this attempt did not make. There is no numerical confirmation or refutation of the paper's central downstream relationship.

5 Figure attempts

These pages identify every main and supplementary figure. A generated image can constitute a partial attempt while failing to reproduce the intended scientific quantity. In particular, a missing-data notice is never counted as a reproduced panel. The detailed ledger following these pages gives panel-level quantities, parameters, exact commands, and limitations.

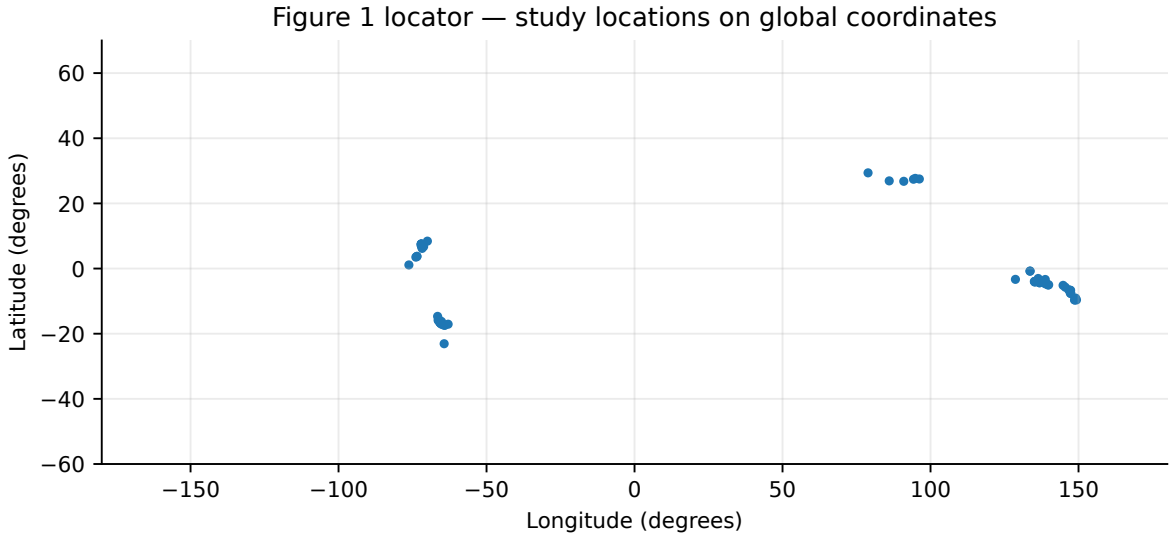


Figure 3: Manuscript Fig. 1 locator attempt. The global coordinates are generated from the permitted metadata; continental outlines and the original inset design are absent. This is a partial geographic context plot.

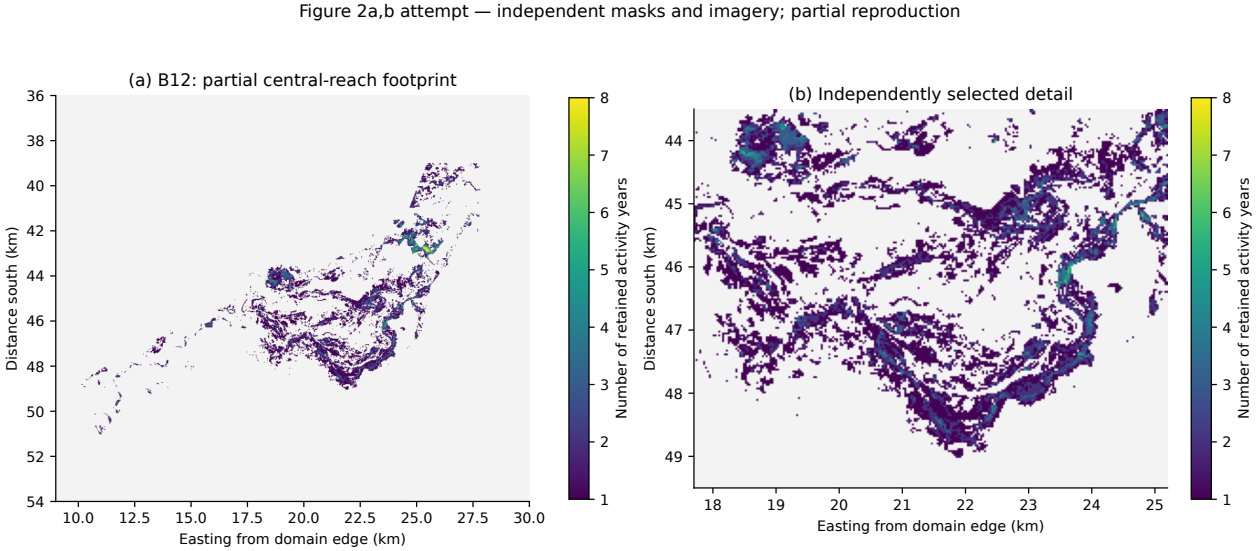


Figure 4: Manuscript Fig. 2a,b attempt. A valid generated footprint, if available, uses only the independent imagery and annotations. An absence notice identifies a missing scientific result; it must not be interpreted as a zero-area event.

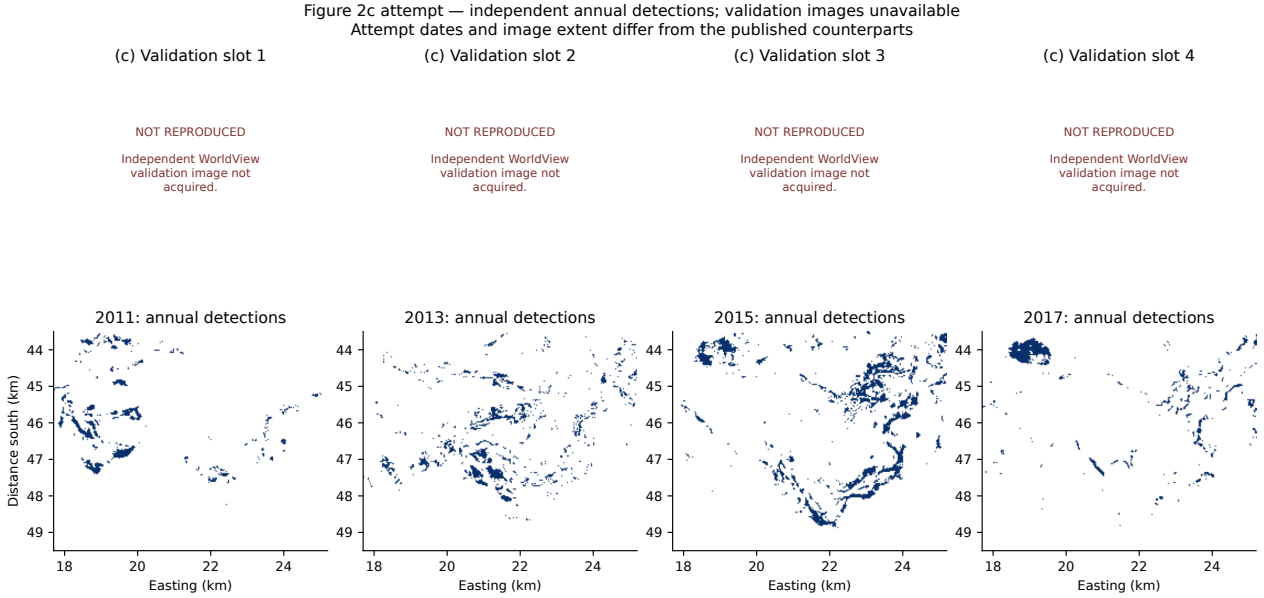


Figure 5: Manuscript Fig. 2c attempt, with four upper validation slots and four lower annual-detection slots. Independently acquired WorldView observations are unavailable. Attempt years 2011, 2013, 2015, and 2017 are explicit choices, not the published annual counterparts 2007, 2010, 2013, and 2017. The latter were identified only during post hoc comparison and did not change the inputs.

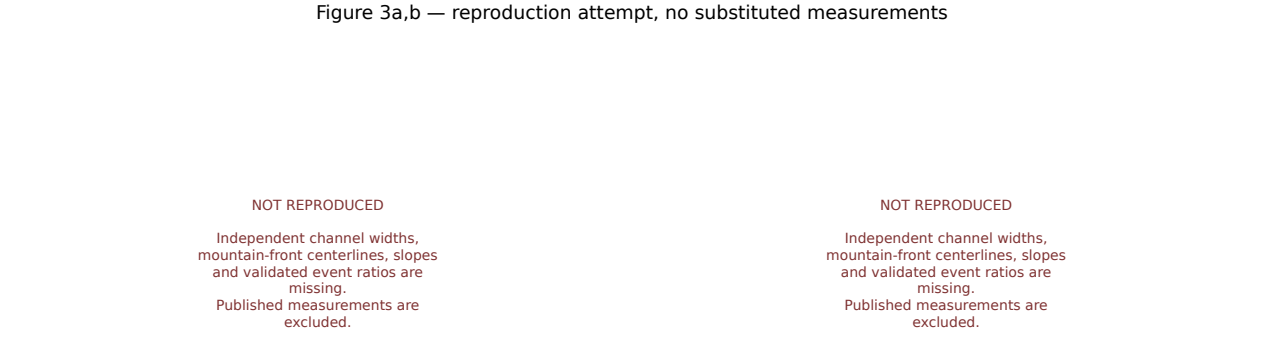


Figure 6: Manuscript Fig. 3a,b accounting panels. Missing independent channel geometry and complete event ratios prevent either statistical comparison. No published point coordinates or synthetic observations are plotted.

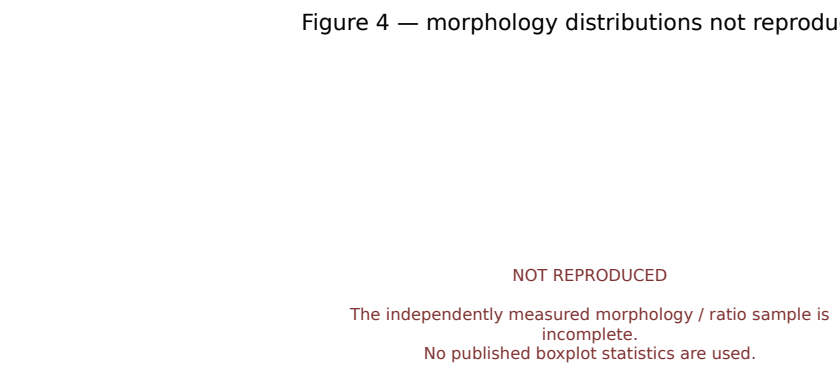


Figure 7: Manuscript Fig. 4 accounting panel. The independently measured morphology–style-ratio sample is empty; no boxplot can be reproduced.

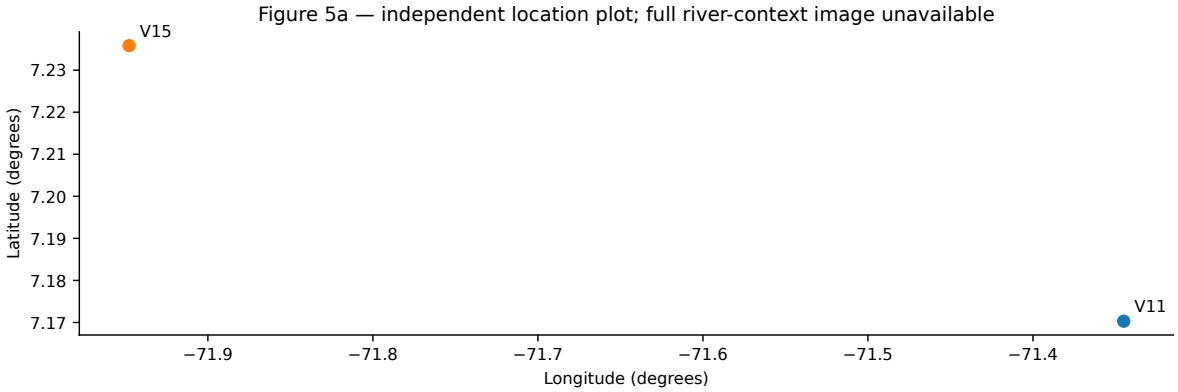


Figure 8: Manuscript Fig. 5a partial attempt. The independently plotted site positions do not replace the paper's river-context mosaic or measure along-channel distance.

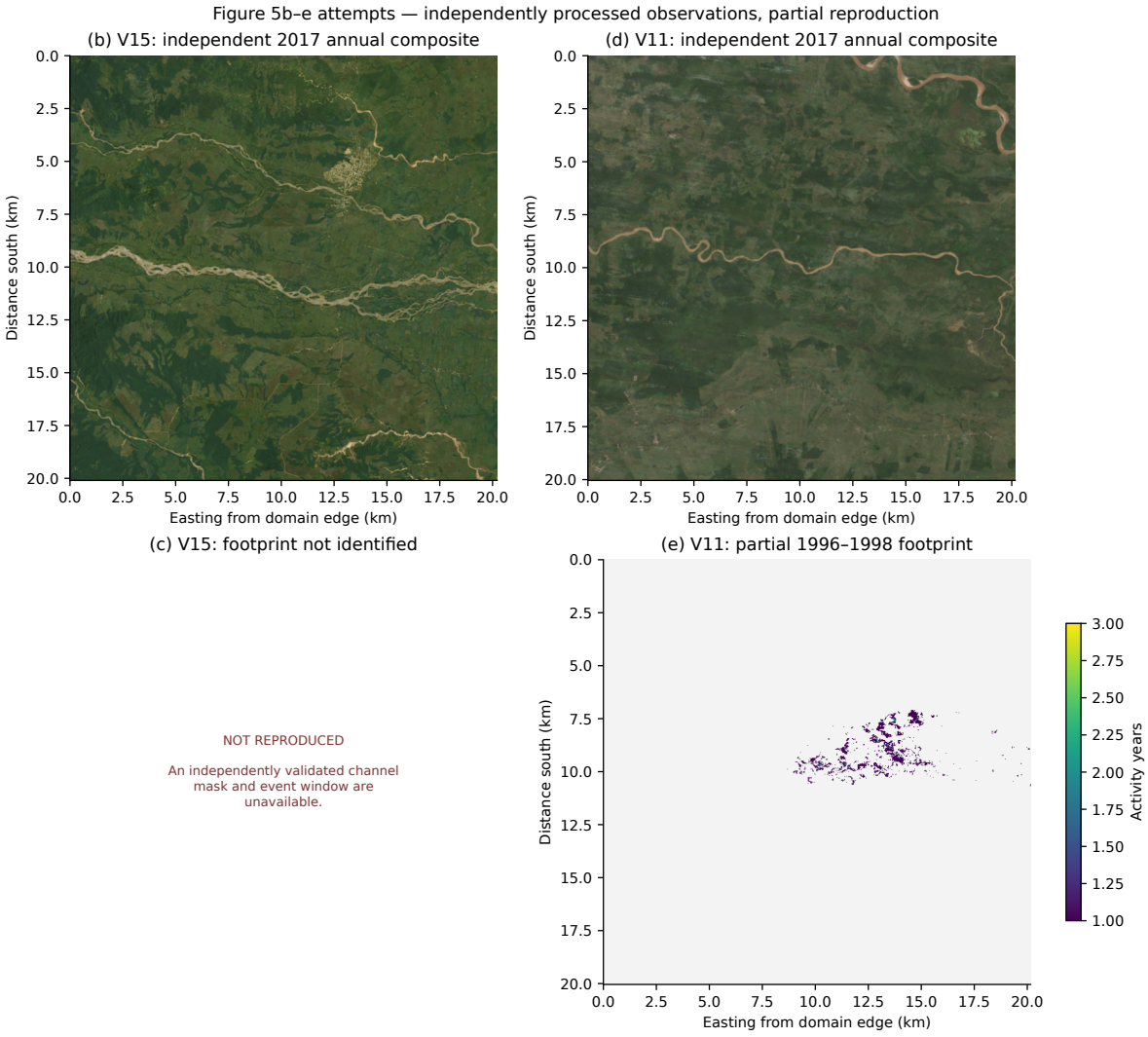


Figure 9: Manuscript Fig. 5b–e attempts. Top: independent 2017 annual RGB composites, displayed with the same fixed stretch. These differ from the paper's 2019 Google Earth context images. Bottom left: V15 event not identified. Bottom right: V11's partial 1996–1998 duration fingerprint, using the selected brightness signature and an independently drawn polygon. The event crosses the eastern domain edge; this is not a complete event or a validated style ratio. Colors count retained years of spectral detection.

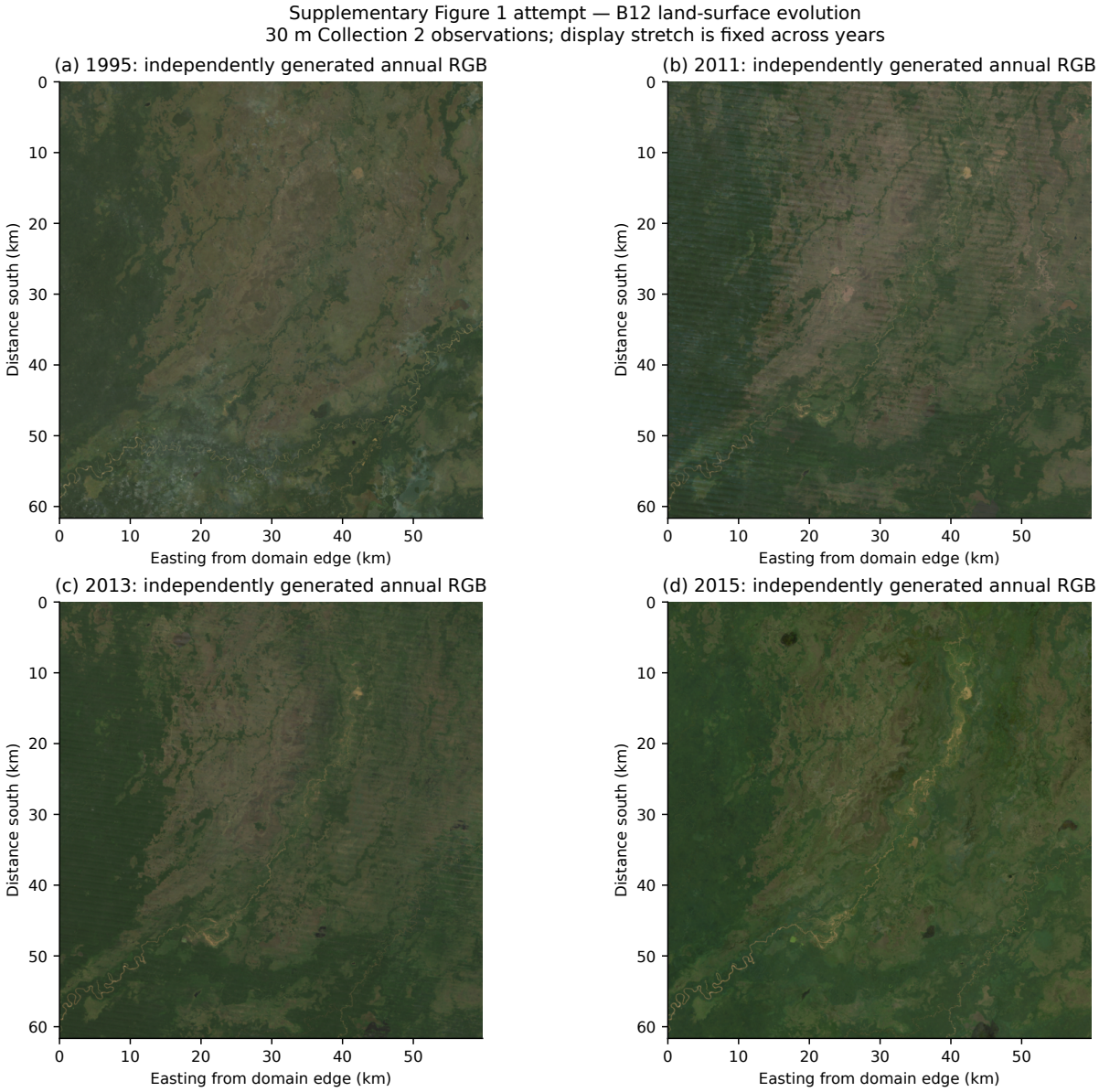


Figure 10: Supplementary Fig. 1a–d attempts: independent B12 annual RGB composites for 1995, 2011, 2013, and 2015 on the larger 30-m domain. The stretch divides reflectance by 0.35 and applies exponent 1/1.6; missing pixels are gray. These are partial observations with independently chosen extent, not copies of the publication's panels.

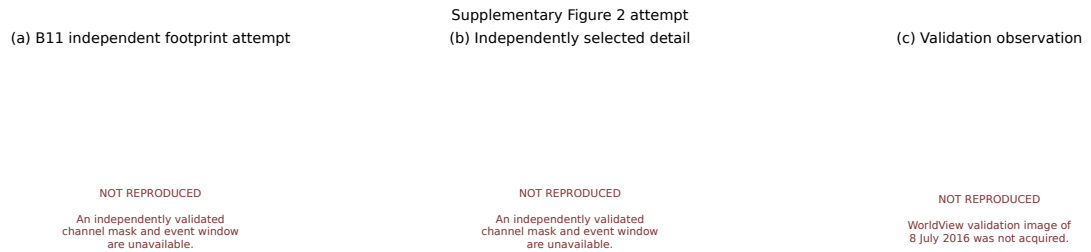


Figure 11: Supplementary Fig. 2a–c attempts. A complete B11 fingerprint and the independent WorldView validation observation of 8 July 2016 remain unavailable. The missing-result notices are not reproduced scientific panels.

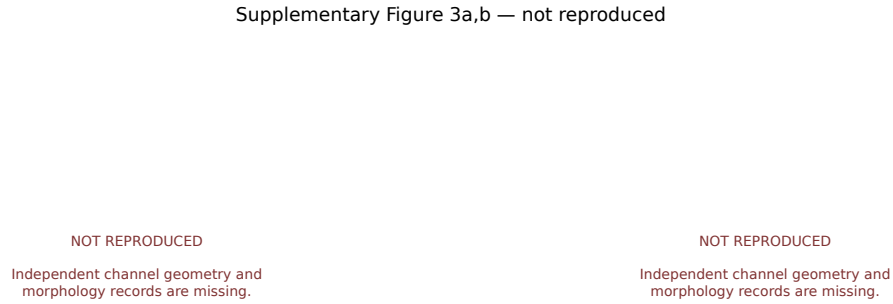


Figure 12: Supplementary Fig. 3a,b accounting panels. Independent slope, normalized distance, and morphology measurements remain incomplete.

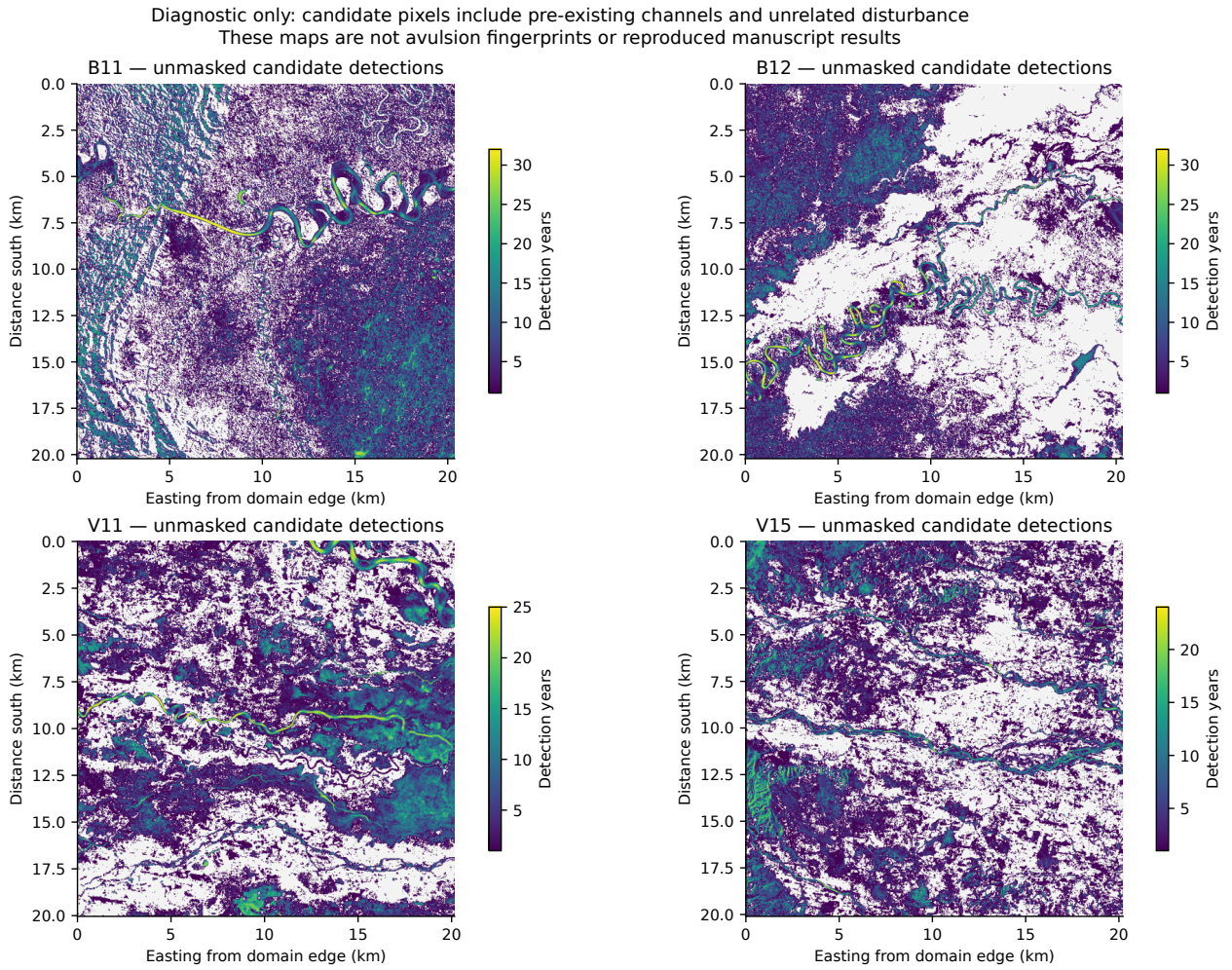


Figure 13: Additional diagnostic, not a manuscript reproduction. Unmasked wetness-signature detection-year counts include pre-existing rivers, clouds that escape the alternate screen, and unrelated disturbance. The accumulation uses all available years and is not an avulsion event footprint.

6 Panel-by-panel record

"Independently reproduced" is restricted here to the geographic site distribution in Fig. 1a–d, generated by this implementation from permitted location metadata and independent terrain. It does not denote a reproduced model prediction or identical cartography. "Partially reproduced" denotes generated observations with stated departures or missing components. "Not reproduced" denotes absence of the panel's scientific quantity. The machine-readable version is `data/results/panel_record.csv`. Commands are relative to the repository root; the acquisition and build prerequisites are given below. A missing-annotation command records the unsatisfied prerequisite and is not run by the successful default procedure.

Panel 1a — independently reproduced

Site distribution: Northern Andes (16). Main Fig. 1 caption; permitted site coordinates.

Parameters: 63 coordinates overall; independent zoom-7 terrain resampled to 600 x 800; B15 included.

Data: `build/avulsion catalog config/sites.csv data/results; .runtime/py311/bin/python scripts/terrain.py`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig1`.

Reproduces the geographic site distribution, not pixel-identical cartography. Background source, map bounds and label placement differ. This is an observational map, not a model result.

Panel 1b — independently reproduced

Site distribution: Central Andes (14). Main Fig. 1 caption; permitted site coordinates.

Parameters: 63 coordinates overall; independent zoom-7 terrain resampled to 600 x 800; B15 included.

Data: `build/avulsion catalog config/sites.csv data/results; .runtime/py311/bin/python scripts/terrain.py`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig1`.

Reproduces the geographic site distribution, not pixel-identical cartography. Background source, map bounds and label placement differ. This is an observational map, not a model result.

Panel 1c — independently reproduced

Site distribution: Himalaya (6). Main Fig. 1 caption; permitted site coordinates.

Parameters: 63 coordinates overall; independent zoom-7 terrain resampled to 600 x 800; B15 included.

Data: `build/avulsion catalog config/sites.csv data/results; .runtime/py311/bin/python scripts/terrain.py`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig1`.

Reproduces the geographic site distribution, not pixel-identical cartography. Background source, map bounds and label placement differ. This is an observational map, not a model result.

Panel 1d — independently reproduced

Site distribution: New Guinea (27). Main Fig. 1 caption; permitted site coordinates.

Parameters: 63 coordinates overall; independent zoom-7 terrain resampled to 600 x 800; B15 included.

Data: `build/avulsion catalog config/sites.csv data/results; .runtime/py311/bin/python scripts/terrain.py`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig1`.

Reproduces the geographic site distribution, not pixel-identical cartography. Background source, map bounds and label placement differ. This is an observational map, not a model result.

Panel 1 locator — partially reproduced

Global study-region locations. Main Fig. 1.

Parameters: Same coordinates; rectangular longitude/latitude axes.

Data: `build/avulsion catalog config/sites.csv data/results`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig1`.

Global positions generated independently; continental outline and original inset design absent.

Panel 2a — partially reproduced

Cumulative B12 disturbance footprint. Main Fig. 2 caption; Methods fingerprinting; Supplementary Table 2.

Parameters: 30 m; B90/G50 and W75/G50; independent rectangular domain and manual polygon if available.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide; build/avulsion measure data/derived/B12_wide config/annotations/B12_wide/polygons.csv data/derived/B12_wide/event`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

Published spatial pattern and parent/new-channel relationship are not recovered. This is a partial spectral-method attempt, not a verified reconstruction of the published event. See docs/published-comparison.md.

Panel 2b — partially reproduced

Detail of B12 disturbance footprint. Main Fig. 2 caption; Methods fingerprinting; Supplementary Table 2.

Parameters: 30 m; B90/G50 and W75/G50; independent rectangular domain and manual polygon if available.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide; build/avulsion measure data/derived/B12_wide config/annotations/B12_wide/polygons.csv data/derived/B12_wide/event`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

Published spatial pattern and parent/new-channel relationship are not recovered. This is a partial spectral-method attempt, not a verified reconstruction of the published event. See docs/published-comparison.md.

Panel 2c upper 1 — not reproduced

High-resolution validation observation. Main Fig. 2 caption.

Parameters: WorldView imagery; published dates recorded only for post hoc comparison.

Data: `No data command: independently licensed validation image unavailable`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

An explicit absence notice occupies this slot. No copied validation image is displayed.

Panel 2c lower 1 — partially reproduced

Annual B12 detection counterpart (attempt year 2011). Main Fig. 2 caption; Methods; Supplementary Table 2.

Parameters: 30 m annual median; selected signature; four attempt years are independently declared.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide; build/avulsion measure data/derived/B12_wide config/annotations/B12_wide/polygons.csv data/derived/B12_wide/event`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

Published annual counterparts are 2007, 2010, 2013, 2017, identified only during post hoc assessment; our declared years remain unchanged. High-resolution counterparts unavailable; no pixel agreement score can be computed.

Panel 2c upper 2 — not reproduced

High-resolution validation observation. Main Fig. 2 caption.

Parameters: WorldView imagery; published dates recorded only for post hoc comparison.

Data: `No data command: independently licensed validation image unavailable`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

An explicit absence notice occupies this slot. No copied validation image is displayed.

Panel 2c lower 2 — partially reproduced

Annual B12 detection counterpart (attempt year 2013). Main Fig. 2 caption; Methods; Supplementary Table 2.

Parameters: 30 m annual median; selected signature; four attempt years are independently declared.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide; build/avulsion measure data/derived/B12_wide config/annotations/B12_wide/polygons.csv data/derived/B12_wide/event`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

Published annual counterparts are 2007, 2010, 2013, 2017, identified only during post hoc assessment; our declared years remain unchanged. High-resolution counterparts unavailable; no pixel agreement score can be computed.

Panel 2c upper 3 — not reproduced

High-resolution validation observation. Main Fig. 2 caption.

Parameters: WorldView imagery; published dates recorded only for post hoc comparison.

Data: `No data command: independently licensed validation image unavailable`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

An explicit absence notice occupies this slot. No copied validation image is displayed.

Panel 2c lower 3 — partially reproduced

Annual B12 detection counterpart (attempt year 2015). Main Fig. 2 caption; Methods; Supplementary Table 2.

Parameters: 30 m annual median; selected signature; four attempt years are independently declared.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide; build/avulsion measure data/derived/B12_wide config/annotations/B12_wide/polygons.csv data/derived/B12_wide/event`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

Published annual counterparts are 2007, 2010, 2013, 2017, identified only during post hoc assessment; our declared years remain unchanged. High-resolution counterparts unavailable; no pixel agreement score can be computed.

Panel 2c upper 4 — not reproduced

High-resolution validation observation. Main Fig. 2 caption.

Parameters: WorldView imagery; published dates recorded only for post hoc comparison.

Data: `No data command: independently licensed validation image unavailable`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

An explicit absence notice occupies this slot. No copied validation image is displayed.

Panel 2c lower 4 — partially reproduced

Annual B12 detection counterpart (attempt year 2017). Main Fig. 2 caption; Methods; Supplementary Table 2.

Parameters: 30 m annual median; selected signature; four attempt years are independently declared.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide; build/avulsion measure data/derived/B12_wide config/annotations/B12_wide/polygons.csv data/derived/B12_wide/event`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig2`.

Published annual counterparts are 2007, 2010, 2013, 2017, identified only during post hoc assessment; our declared years remain unchanged. High-resolution counterparts unavailable; no pixel agreement score can be computed.

Panel 3a — not reproduced

Style ratio versus distance / width. Main Figs. 3, 4; Supplementary Fig. 3; Methods channel measurements.

Parameters: D/w; elevation drop / reach length; three morphology classes; D/w $>$ 60 fit subset; range whiskers.

Data: `build/avulsion statistics config/sites.csv config/channel_measurements.csv data/derived data/results`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig3`.

No complete independent event-plus-channel-geometry record exists. All 63 sites are audited; empty results stay empty. Statistical arithmetic is tested on analytical fixtures, which are not plotted as study data.

Panel 3b — not reproduced

Style ratio versus channel-belt slope. Main Figs. 3, 4; Supplementary Fig. 3; Methods channel measurements.

Parameters: D/w; elevation drop / reach length; three morphology classes; D/w $>$ 60 fit subset; range whiskers.

Data: `build/avulsion statistics config/sites.csv config/channel_measurements.csv data/derived data/results`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig3`.

No complete independent event-plus-channel-geometry record exists. All 63 sites are audited; empty results stay empty. Statistical arithmetic is tested on analytical fixtures, which are not plotted as study data.

Panel 4 — not reproduced

Style-ratio distributions by morphology. Main Figs. 3, 4; Supplementary Fig. 3; Methods channel measurements.

Parameters: D/w; elevation drop / reach length; three morphology classes; D/w > 60 fit subset; range whiskers.

Data: `build/avulsion statistics config/sites.csv config/channel_measurements.csv data/derived data/results`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig4`.

No complete independent event-plus-channel-geometry record exists. All 63 sites are audited; empty results stay empty. Statistical arithmetic is tested on analytical fixtures, which are not plotted as study data.

Panel S3a — not reproduced

Slope distributions by morphology. Main Figs. 3, 4; Supplementary Fig. 3; Methods channel measurements.

Parameters: D/w; elevation drop / reach length; three morphology classes; D/w > 60 fit subset; range whiskers.

Data: `build/avulsion statistics config/sites.csv config/channel_measurements.csv data/derived data/results`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp3`.

No complete independent event-plus-channel-geometry record exists. All 63 sites are audited; empty results stay empty. Statistical arithmetic is tested on analytical fixtures, which are not plotted as study data.

Panel S3b — not reproduced

Distance / width distributions by morphology. Main Figs. 3, 4; Supplementary Fig. 3; Methods channel measurements.

Parameters: D/w; elevation drop / reach length; three morphology classes; D/w > 60 fit subset; range whiskers.

Data: `build/avulsion statistics config/sites.csv config/channel_measurements.csv data/derived data/results`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp3`.

No complete independent event-plus-channel-geometry record exists. All 63 sites are audited; empty results stay empty. Statistical arithmetic is tested on analytical fixtures, which are not plotted as study data.

Panel 5a — partially reproduced

Sarare regional river context. Main Fig. 5 caption.

Parameters: V11 and V15 permitted coordinates.

Data: `build/avulsion catalog config/sites.csv data/results`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig5`.

Location plot only; full river-context satellite mosaic and a traced connecting channel are absent.

Panel 5b — partially reproduced

V15 satellite context. Main Fig. 5 caption; Methods annual composites.

Parameters: Independent 2017 annual median; 30 m; QA screen; fixed display stretch.

Data: `build/avulsion analyze data/raw/V15 data/derived/V15`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig5`.

Paper uses 2019 Google Earth context; our independent 2017 Landsat observation has different timing and resolution.

Panel 5c — not reproduced

V15 disturbance footprint. Main Fig. 5 caption; Methods fingerprinting.

Parameters: 30 m annual candidates; event not uniquely identified.

Data: `build/avulsion analyze data/raw/V15 data/derived/V15`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig5`.

Braided-channel migration and a specific avulsion could not be distinguished confidently from these observations; no event polygon was invented.

Panel 5d — partially reproduced

V11 satellite context. Main Fig. 5 caption; Methods annual composites.

Parameters: Independent 2017 annual median; 30 m; QA screen; fixed display stretch.

Data: `build/avulsion analyze data/raw/V11 data/derived/V11`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig5`.

Paper uses 2019 Google Earth context; our independent 2017 Landsat observation has different timing and resolution.

Panel 5e — partially reproduced

V11 disturbance footprint. Main Fig. 5 caption; Methods fingerprinting.

Parameters: V11: parent 1986; event observation 1996–1998; partial manual mask; selected signature.

Data: `build/avulsion analyze data/raw/V11 data/derived/V11; build/avulsion measure data/derived/V11 config/annotations/V11/polygons.csv data/derived/V11/event`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure fig5`.

V11 has unequal windows and disturbance crossing the domain edge; its diagnostic ratio is not a full-event style estimate.

Panel S1a — partially reproduced

B12 land surface in 1995. Supplementary Fig. 1 caption; Methods annual compositing.

Parameters: 1995; 30 m; independent Collection 2 annual RGB; fixed reflectance / 0.35 display stretch and gamma 1/1.6.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp1`.

Same named site and years; independently chosen domain, different processing and no verified match to original image footprint.

Panel S1b — partially reproduced

B12 land surface in 2011. Supplementary Fig. 1 caption; Methods annual compositing.

Parameters: 2011; 30 m; independent Collection 2 annual RGB; fixed reflectance / 0.35 display stretch and gamma 1/1.6.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp1`.

Same named site and years; independently chosen domain, different processing and no verified match to original image footprint.

Panel S1c — partially reproduced

B12 land surface in 2013. Supplementary Fig. 1 caption; Methods annual compositing.

Parameters: 2013; 30 m; independent Collection 2 annual RGB; fixed reflectance / 0.35 display stretch and gamma 1/1.6.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp1`.

Same named site and years; independently chosen domain, different processing and no verified match to original image footprint.

Panel S1d — partially reproduced

B12 land surface in 2015. Supplementary Fig. 1 caption; Methods annual compositing.

Parameters: 2015; 30 m; independent Collection 2 annual RGB; fixed reflectance / 0.35 display stretch and gamma 1/1.6.

Data: `build/avulsion analyze data/raw/B12_wide data/derived/B12_wide`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp1`.

Same named site and years; independently chosen domain, different processing and no verified match to original image footprint.

Panel S2a — not reproduced

B11 fingerprint. Supplementary Fig. 2; Methods fingerprinting.

Parameters: 30 m; 1986–2017 archive; both published spectral signatures.

Data: `build/avulsion analyze data/raw/B11 data/derived/B11`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp2`.

Annual composites and candidate masks were generated, but a unique complete event mask and timing were not established.

Panel S2b — not reproduced

B11 fingerprint detail. Supplementary Fig. 2; Methods fingerprinting.

Parameters: 30 m; 1986–2017 archive; both published spectral signatures.

Data: `build/avulsion analyze data/raw/B11 data/derived/B11`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp2`.

Annual composites and candidate masks were generated, but a unique complete event mask and timing were not established.

Panel S2c — not reproduced

B11 high-resolution validation image. Supplementary Fig. 2 caption.

Parameters: WorldView observation dated 8 July 2016.

Data: `No data command: independently licensed validation image unavailable`.

Plot: `.runtime/py311/bin/python scripts/plot.py --figure supp2`.

No validation image or copied manuscript panel is substituted.

7 Rebuilding and inspecting the result

All files, caches, Python environments, compilation outputs, temporary downloads, and TeX caches reside under the selected project directory on `/data/data2/`. No project file is saved to a home directory. GitHub CLI reads the user's existing authentication. Publication uses HTTPS under the verified account `RianMontagh`.

Requirements are a C++17 compiler, CMake, Python 3.11 or newer with venv, and pdfLaTeX with the standard packages used by this source. The pinned Python dependencies are in `requirements.txt`. From a checkout placed under a permitted data directory, the complete procedure is

```
./rebuild.sh --download
```

It builds and tests the C++ executable; retrieves the frozen agency scene catalogs and terrain inputs; computes annual composites and spectral masks; measures available independent polygons; audits all 63 statistical records; generates plots and the panel ledger; archives saved arrays losslessly; and compiles this PDF twice to resolve references. Network acquisition may take tens of minutes and requires substantial disk space. Failed downloads are recorded rather than concealed. Inputs must not be replaced with author masks or imagery.

Once the raw cache exists, the verified computation can be repeated with

```
./rebuild.sh --offline
```

This is the same build, analysis, plotting, and report procedure with network retrieval skipped. It recomputes results from the cached agency arrays, not from plots. To preserve the run log and include the actual completed outcome in the report, use `python3.11 scripts/run_rebuild.py --offline`. The documented `--download` procedure completed with exit code 0 in 1063.7 s. `data/results/rebuild_outcome.json` records the command, platform, and tool versions. The executed run, command, versions, and exit status are recorded in `data/results/rebuild_outcome.json` and its accompanying log. The PDF can also be rebuilt from archived generated arrays by unpacking them with `scripts/pack_outputs.py--unpack` and following the figure-only procedure in the README. That shortcut does not rerun the scientific analysis.

The binary raster outputs use row-major little-endian arrays; filenames identify float32, uint16, or uint8 type. Grid files specify dimensions, 30-m pixel size, upper-left coordinates, and EPSG code. Tasseled-cap and RGB rasters are band-major. Annual CSVs report sensor selection, sample counts, thresholds, coverage, and both detection areas. All generated arrays are included in lossless archives; provider raw scene cutouts can be regenerated from frozen unsigned URLs and hashes. `data/results/artifact_sha256.json` records hashes of the deliverable artifacts. The report PDF is compiled from this LaTeX source and generated figures, never from manuscript image files.

8 Assessment

The spectral and counting operations are comparatively straightforward to implement and verify. They require no undocumented physical constitutive law. Independent raw observations can be acquired without the researchers' code or processed data. Geographic site mapping and annual reflectance imaging therefore provide reproducible intermediate results.

The dominant limitation is recovering the observation design: event start and end, the complete disturbed floodplain, the preceding channel, unrelated water or exposed sediment, mountain-front position, representative width, local elevation profile, and thread-count classification. These determine both axes of the central statistical figures. Annual composites also hide within-year timing, cloudy intervals hide event development, and 30-m pixels cannot resolve narrow channels or subcanopy disturbance. Spectral exposure is a proxy for floodplain disturbance, not a direct measurement of sediment volume or cause.

This attempt produces an independently auditable computation and a partial set of observations. It does not recover the paper's 63-event result. The numerical tests establish correct implementation of the declared operations; they cannot compensate for absent independent observations. A subsequent test using the authors' processed images and masks would address a different question: agreement conditional on their observational choices. That test has not been performed here.

References

[1] Valenza, J. M., Edmonds, D. A., Hwang, T., and Roy, S. (2020). Downstream changes in river avulsion style are related to channel morphology. *Nature Communications*, 11, 2116. https://doi.org/10.1038/s41467-020-15859-9.

[2] Valenza et al. (2020). Supplementary information, especially Table 2, methodological notes, and figure captions. Available from the article's supplementary-information links. Only the separately permitted site identifiers and coordinates are retained from the location table.

[3] U.S. Geological Survey. How do I use a scale factor with Landsat Level-2 science products? https://www.usgs.gov/faqs/how-do-i-use-a-scale-factor-landsat-level-2-science-products.

[4] U.S. Geological Survey. Landsat Collection 2 quality assessment bands. https://www.usgs.gov/landsat-missions/landsat-collection-2-quality-assessment-bands.

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